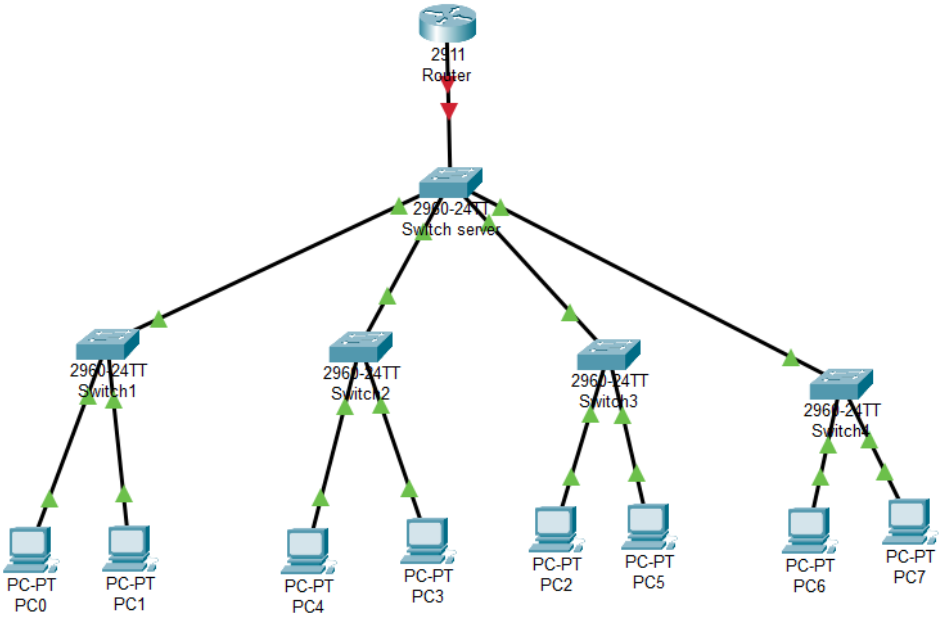
1) Разместим оборудование и выполним базовое подключение:



2) Выполним базовую настройку коммутаторов:

Switch server:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW-Server
SW-Server(config)#enable secret SwitchPass123
SW-Server(config)#service password-encryption
SW-Server(config)#line console 0
SW-Server(config-line)#password console123
SW-Server(config-line)#login
SW-Server(config-line)#line vty 0 15
SW-Server(config-line)#password vty123
SW-Server(config-line)#login
SW-Server(config-line)#exit
SW-Server(config)#end
SW-Server#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
SW-Server#
```

Switch1:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#enable secret SwitchPass123
SW1(config)#service password-encryption
SW1(config)#line console 0
SW1(config-line)#password console123
SW1(config-line)#login
SW1(config-line)#exit
SW1(config)#line vty 0 15
SW1(config-line)#password vty123
SW1(config-line)#login
SW1(config-line)#exit
SW1(config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Switch2:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW2
SW2(config)#enable secret SwitchPass123
SW2(config)#service password-encryption
SW2(config)#line console 0
SW2(config-line)#password console123
SW2(config-line)#login
SW2(config-line)#exit
SW2(config)#line vty 0 15
SW2(config-line)#password vty123
SW2(config-line)#login
SW2(config-line)#exit
SW2(config)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
SW2#
```

Switch3:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW3
SW3(config)#enable secret SwitchPass123
SW3(config)#service password-encryption
SW3(config)#line console 0
SW3(config-line)#password vty123
SW3(config-line)#password console123
SW3(config-line)#login
SW3(config-line)#exit
SW3(config)#line vty 0 15
SW3(config-line)#password vty123
SW3(config-line)#login
SW3(config-line)#exit
SW3(config)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
SW3#
SW3#exit
```

Switch4:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW4
SW4(config)#enable secret SwitchPass123
SW4(config)#service password-encryption
SW4(config)#line console 0
SW4(config-line)#password console123
SW4(config-line)#login
SW4(config-line)#exit
SW4(config)#line vty 0 15
SW4(config-line)#password vty123
SW4(config-line)#login
SW4(config-line)#exit
SW4(config)#end
SW4#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
SW4#
```

3) Настройка VTP

На Switch Server (VTP Server):

```
User Access Verification

Password:

SW-Server>enable
Password:
SW-Server#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-Server(config)#vtp domain CompanyNet
Changing VTP domain name from NULL to CompanyNet
SW-Server(config)#vtp mode server
Device mode already VTP SERVER.
SW-Server(config)#vtp password VTPpass123
Setting device VLAN database password to VTPpass123
SW-Server(config)#
```

На Switch1:

```
SW1(config)#vtp domain CompanyNet
Domain name already set to CompanyNet.
SW1(config)#vtp mode client
Device mode already VTP CLIENT.
SW1(config)#vtp password VTPpass123
Password already set to VTPpass123
```

На Switch2:

```
SW2(config)#vtp domain CompanyNet
Changing VTP domain name from NULL to CompanyNet
SW2(config)#vtp mode client
Setting device to VTP CLIENT mode.
SW2(config)#vtp password VTPpass123
Setting device VLAN database password to VTPpass123
```

На Switch3:

```
SW3(config)#vtp domain CompanyNet
Changing VTP domain name from NULL to CompanyNet
SW3(config)#vtp mode client
Setting device to VTP CLIENT mode.
SW3(config)#vtp password VTPpass123
Setting device VLAN database password to VTPpass123
SW3(config)#exit
```

На Switch4:

```
SW4(config)#vtp domain CompanyNet
Changing VTP domain name from NULL to CompanyNet
SW4(config)#vtp mode client
Setting device to VTP CLIENT mode.
SW4(config)#vtp password VTPpass123
Setting device VLAN database password to VTPpass123
```

4) Создание VLAN на сервере VTP

```
SW-Server>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
24   Developers                       active
38   Management                       active
62   Sales                            active
86   Guest                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

5) Настройка Trunk-портов

Ha Switch server

```
SW-Server(config)#interface range fa0/1-5
SW-Server(config-if-range)#switchport mode trunk



SW-Server(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/4 (86), with SW3
FastEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/5 (86), with SW4
FastEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (86), with SW2
FastEthernet0/1 (1).

SW-Server(config-if-range)#switchport trunk native vlan 86
SW-Server(config-if-range)#no shutdown
```

Ha Switch1:

```
SW1(config)#interface fa0/1
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport trunk native vlan 86
SW1(config-if)#no shutdown
SW1(config-if)#exit
```

Ha Switch2:

```
SW2(config-if)#switchport mode trunk
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW-Server
FastEthernet0/3 (86).

SW2(config-if)#switchport trunk native vlan 86
SW2(config-if)#%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/1 on VLAN0086. Port
consistency restored.

%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/1 on VLAN0001. Port consistency
restored.


SW2(config-if)#no shutdown
SW2(config-if)#exit
```

Ha Switch3:

```
SW3(config-if)#switchport mode trunk
SW3(config-if)#switchport trunk native vlan 86
SW3(config-if)#%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/1 on VLAN0086. Port
consistency restored.

%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/1 on VLAN0001. Port consistency
restored.

SW3(config-if)#no shutdown
```

Ha Switch4:

```
SW4(config-if)#switchport mode trunk

SW4(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%SPANTREE-2-RECV_PVID_ERR: Received BPDU with inconsistent peer vlan id 86 on FastEthernet0/1
VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/1 on VLAN0001. Inconsistent local vlan.


SW4(config-if)#switchport trunk native vlan 86
SW4(config-if)#%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/1 on VLAN0086. Port
consistency restored.

%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/1 on VLAN0001. Port consistency
restored.

no shutdown
SW4(config-if)#no shutdown
```

6) Настройка маршрутизатора (Router-on-a-Stick)

```
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0.24
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.24, changed state to up

Router(config-subif)#encapsulation dot1Q 24
Router(config-subif)#ip address 192.168.24.1
% Incomplete command.
Router(config-subif)#ip address 192.168.24.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface GigabitEthernet0/0.62
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.62, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.62, changed state to up

Router(config-subif)#encapsulation dot1Q 62
Router(config-subif)#ip address 192.168.62.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface GigabitEthernet0/0.86
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.86, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.86, changed state to up

Router(config-subif)#encapsulation dot1Q 86
Router(config-subif)#ip address 192.168.86.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface GigabitEthernet0/0.38
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.38, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.38, changed state to up

Router(config-subif)#ip address 192.168.38.1 255.255.255.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

Router(config-subif)#encapsulation dot1Q 38
Router(config-subif)#ip address 192.168.38.1 255.255.255.0
Router(config-subif)#exit
```

7) Настройка Access-портов4

На Switch1 (VLAN 24):

```
SW1>enable
Password:
Password:
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 24
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#interface fa0/3
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 24
SW1(config-if)#no shutdown
SW1(config-if)#exit
```

На Switch2 (VLAN 62):

```
SW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#interface fa0/2
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 62
SW2(config-if)#no shutdown
SW2(config-if)#exit
SW2(config)#interface fa0/3
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 62
SW2(config-if)#no shutdown
SW2(config-if)#exit
```

На Switch3 (VLAN 86):

```
SW3>enable
Password:
SW3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#interface fa0/2
SW3(config-if)#switchport mode access
SW3(config-if)#switchport access vlan 86
SW3(config-if)#no shutdown
SW3(config-if)#exit
SW3(config)#interface fa0/3
SW3(config-if)#switchport mode access
SW3(config-if)#switchport access vlan 86
SW3(config-if)#no shutdown
SW3(config-if)#exit
```

На Switch4 (VLAN 38):

```
SW4>enable
Password:
SW4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW4(config)#interface fa0/2
SW4(config-if)#switchport mode access
SW4(config-if)#switchport access vlan 38
SW4(config-if)#no shutdown
SW4(config-if)#exit
SW4(config)#interface fa0/3
SW4(config-if)#switchport mode access
SW4(config-if)#switchport access vlan 38
SW4(config-if)#no shutdown
SW4(config-if)#exit
```

8) Настройка DHCP-сервера на маршрутизаторе

Исключение адресов шлюзов и резервных адресов:

```
Router(config)#ip dhcp excluded-address 192.168.24.1 192.168.24.10
Router(config)#ip dhcp excluded-address 192.168.62.1 192.168.62.10
Router(config)#ip dhcp excluded-address 192.168.86.1 192.168.86.10
Router(config)#ip dhcp excluded-address 192.168.38.1 192.168.38.10
```

Создание пулов DHCP:

```
Router(config)#ip dhcp pool DEVELOPERS
Router(dhcp-config)#network 192.168.24.1
% Incomplete command.
Router(dhcp-config)#network 192.168.24.1 255.255.255.0
Router(dhcp-config)#exit
Router(config)#
Router(config)#
Router(config)#ip dhcp pool DEVELOPERS
Router(dhcp-config)#network 192.168.24.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.24.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool SALES
Router(dhcp-config)#network 192.168.62.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.62.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool GUEST
Router(dhcp-config)#network 192.168.86.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.86.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool MANAGEMENT
Router(dhcp-config)#network 192.168.38.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.38.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
```

9) Проверка

На Switch server:

```
SW-Server>show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : CompanyNet
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0001.435C.4400
Configuration last modified by 0.0.0.0 at 3-1-93 06:56:25
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 9
Configuration Revision            : 8
MD5 digest                        : 0x5E 0x8D 0x92 0x42 0x26 0xB5 0xD2 0xCE
                                    0x11 0xC9 0x7D 0x8B 0x30 0xC6 0x12 0x8A
```

На Swith1:

```
SW1>show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : CompanyNet
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 00D0.97D7.3400
Configuration last modified by 0.0.0.0 at 3-1-93 06:56:25

Feature VLAN :
--------------
VTP Operating Mode                : Client
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 9
Configuration Revision            : 8
MD5 digest                        : 0x5E 0x8D 0x92 0x42 0x26 0xB5 0xD2 0xCE
                                    0x11 0xC9 0x7D 0x8B 0x30 0xC6 0x12 0x8A
```

На Swith2:

```
SW2>show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : CompanyNet
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0090.2B17.A500
Configuration last modified by 0.0.0.0 at 3-1-93 06:56:25

Feature VLAN :
--------------
VTP Operating Mode                : Client
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 9
Configuration Revision            : 8
MD5 digest                        : 0x5E 0x8D 0x92 0x42 0x26 0xB5 0xD2 0xCE
                                    0x11 0xC9 0x7D 0x8B 0x30 0xC6 0x12 0x8A
```

На Swith3:

```
SW3>show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : CompanyNet
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0001.C92C.4500
Configuration last modified by 0.0.0.0 at 3-1-93 06:56:25

Feature VLAN :
--------------
VTP Operating Mode                : Client
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 9
Configuration Revision            : 8
MD5 digest                        : 0x5E 0x8D 0x92 0x42 0x26 0xB5 0xD2 0xCE
                                    0x11 0xC9 0x7D 0x8B 0x30 0xC6 0x12 0x8A
```

На Swith4:

```
SW4>show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : CompanyNet
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 00D0.FF72.9C00
Configuration last modified by 0.0.0.0 at 3-1-93 06:56:25

Feature VLAN :
--------------
VTP Operating Mode                : Client
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 9
Configuration Revision            : 8
MD5 digest                        : 0x5E 0x8D 0x92 0x42 0x26 0xB5 0xD2 0xCE
                                    0x11 0xC9 0x7D 0x8B 0x30 0xC6 0x12 0x8A
```

Ответы на контрольные вопросы:

1) Роль VTP - централизованное управление VLAN. Создается VLAN на одном коммутаторе (VTP Server), а остальные автоматически получают эти настройки через VTP.
2) Принцип Router-on-a-Stick заключается в том, что один физический порт маршрутизатора подключается к коммутатору и используется для маршрутизации трафика между несколькими VLAN с помощью подынтерфейсов. Для этого используется инкапсуляция 802.1Q, которая позволяет передавать тегированные кадры (с меткой VLAN) по одному каналу. Подынтерфейсы с инкапсуляцией 802.1Q необходимы для того, чтобы маршрутизатор мог различать трафик.
3) Для настройки DHCP на маршрутизаторе Cisco для нескольких VLAN необходимо создать виртуальные интерфейсы для каждого VLAN, назначить каждому IP-адрес и маску, а затем создать и

настроить пулы DHCP для каждой подсети. Исключение IP-адресов необходимо для предотвращения конфликтов, так как эти адреса могут использоваться для статической конфигурации сетевых устройств, таких как серверы или принтеры.

4) Access-порт передает трафик только одного VLAN без тегов и используется для подключения конечных устройств. Trunk-порт передает трафик нескольких VLAN с тегами 802.1Q и используется для соединения между коммутаторами или с маршрутизатором. Native VLAN - это VLAN, трафик которого передается по trunk-порту без тегов. Если native VLAN не совпадает на обоих концах транкового соединения, возникает mismatch, что приводит к ошибкам CDP/STP и возможным утечкам трафика между VLAN. Native VLAN обеспечивает совместимость с оборудованием, не поддерживающим тегирование, и должен быть одинаковым на обоих концах trunk-канала.
5) Выполни команды show vtp status на всех коммутаторах - убедись, что домен, режим и количество VLAN совпадают.
show ip dhcp binding покажет какие адреса были выданы и для каких VLAN. Также проверь что шлюзом указан IP соответствующего subinterface маршрутизатора.